

Differential Attribute Desensitization System for Personal Information Protection

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Abstract—Big data brings much convenience to our daily life, while it also raises concerns to personal information leakage. It is important for data owner to protect sensitive data from being attacked. In this paper, we elaborate on the possible attacks that may be suffered at various stages of personal data life cycle and the corresponding privacy protection methods. Based on these methods, a differential attribute desensitization system (DADS) for personal information protection has been proposed. Data owner can define sensitive attribute level in the DADS. And then DADS can automatically identify sensitive data, and take differentiated data desensitization measures for structured and unstructured multi-attribute information. Experimental results show that DADS can automatically adapt to different types of sensitive information and protect personal sensitive information effectively.

Keywords—Privacy protection; Data anonymization; k-anonymity; Data masking; Named entity recognition

I. INTRODUCTION

With the development of big data technology, the collection, processing and analysis of users' information plays an increasingly important role in enterprises to make profits, which also brings challenges to users' privacy and enterprise security, even national security. Based on a large number of user identity information and behavior information, the data owner can use big data technology to analyze more privacy information or users' behavior trends and rules. Those valuable information can be used for enterprise decision-making and precision marketing, but also can become a weapon for criminals. On the one hand, users' identity information is mostly sensitive. Once leaked or illegally obtained, it will pose a great threat to users' privacy, even life and property security. On the other hand, only flowing data can give full play to its value, and information sharing has become a trend. Therefore, the balance between the protection and use of personal information or identity information has become a hot topic in the industry.

The concept of Data Desensitization was first proposed by Adam and Wortmann in 1989 [1]. With the development of relevant technologies, many algorithms have been proposed to solve the problems related to data desensitization, such as noise disturbance, K - anonymous, Micro polymerization. In terms of industrialization, traditional database manufacturers such as IBM, ORACLE and MICROSOFT also launched corresponding data desensitization products.

Traditional data desensitization methods include substitution, mixed washing, numerical transformation, encryption, occlusion and null value insertion or deletion, etc. Those methods are also widely used in commercial scenarios, which should be selected according to requirements and specific scenarios. With the advent of the big data era, people

pay more attention to data mining. Although the traditional method blocks sensitive data, it also damages the value of data, which brings two problems. Therefore, this paper aims to solve the problem of protecting sensitive information while protecting data value and realize the maximum protection of privacy with the least loss of data accuracy.

In this work, we propose DADS, a differential attribute desensitization system for personal information protection, which can automatically adapt to different types of sensitive information and effectively realize differentiated protection of personal sensitive information.

Our contributions are summarized as follows:

- This paper firstly analyzes and introduces the relevant technologies of data desensitization, and emphasizes the difficulty of differential desensitization between structured data and unstructured data.
- This paper adaptively selects appropriate desensitization algorithm for differential protection, by automatically identifying the degree of personal information sensitivity and the type of the data.
- This paper deeply studies the existing data anonymous publishing algorithm, and proposes a personal information desensitization algorithm for regular expression, id number, address and Chinese name, so that it has less information loss and is more suitable for data mining applications.

The rest of the paper is structured as follows. Section 2 presents the background and related works in the literature; Section 3 introduces privacy protection in each life cycle of data. Section 4 introduces the design and implementation of privacy protection technology supporting data desensitization.

II. RELATED WORK

A. Common Attacks

Privacy preservation aims to prevent unauthorized access to sensitive data. Similar as the crack of cipher codes, if the cost of accessing sensitive data is greater than the intrinsic value of data, the protection is successful. In general, common attacks are as follows:

- Correlation Attacks: Adversaries infer some private information by concatenating certain attributes in a table with the same attributes of a known external table. Table 1 shows a partial diagnosis table of a hospital. The name column cannot be directly published, which will directly expose the user's privacy. However, given a known voter information table, as shown in Table2, by comparing the gender, birthplace and birthday attributes of both tables, it is

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possible to affirm that Zhang San suffers from hepatitis B under a certain confidence degree.

- **Homogeneity Attacks:** If the values of sensitive attributes of the records in an equivalence class are the same, homogenous attacks can be performed. For example, as shown in Table 2, we know the gender, birthplace and birthday of Li Si. In a name-removed visit list of a hospital, there are two records filtered out by a QID. It is impossible to determine which one corresponds to Ms. Li Si, but both diagnoses were recorded as asthma, so Ms. Li Si must have asthma.
- **Background Knowledge Attacks:** The adversary has mastered certain basic personal information of the victim. By comparing the attribute information in the published data table with prior knowledge of the mastery, the privacy information of a victim can be obtained with a higher probability, which leads to privacy leaks.
- **Similarity Attacks:** The value of sensitive values in the same equivalence class is different expressions of similar events, with the same semantics, so the adversary can initially determine the victims generalized privacy information. For example, asthma and tuberculosis can preliminarily confirm that the lungs may have disease.
- **Skewed Attacks:** In order to fulfill the diversity of attribute values in the equivalence class, the distribution of attributes in the equivalence class differs greatly from the overall distribution of attributes in the dataset, thus making the adversaries able to determine user privacy at a higher confidence degree. For example, if the probability of a positive in an equivalence class is greater than 60% in attribute of hepatitis B, the adversary can be more confident to infer that the target user has a higher probability of developing hepatitis B.
- **Attribute Inference Attacks:** Adversaries leverage publicly available datasets to infer user privacy attributes through machine learning algorithms.

TABLE I. DIAGNOSIS TABLE OF A HOSPITAL

Name	Gender	Birthplace	Birthday	Diagnosis
Zhang San	Male	Beijing	1990-01-01	Hepatitis B
Li Si	Female	Beijing	1974-05-03	Asthma
Wang Wu	Female	Beijing	1974-05-03	Asthma
Zhao Liu	Male	Beijing	1985-11-28	Phthisis

TABLE II. VOTER INFORMATION TABLE

Name	Gender	Birthplace	Birthday	Party
Zhang San	Male	Beijing	1990-01-01	Communist Party
Li Si	Female	Beijing	1974-05-03	Communist Party

B. Data Anonymization

Data anonymization refers to the certain anonymization of sensitive data, so that the data remains available while the privacy of the user are preserved. The data can be mainly divided into the following four categories:

- **Explicit Identifier (EID):** Each can identify a specific attribute of an individual explicitly. E.g. ID number, name, etc.
- **Quasi-Identifier (QID):** Multiple QIDs can be combined to identify an individual. Usually connected with known tables.
- **Sensitive Value:** Attributes that are private and that the user does not want others to know. E.g. salary, health status, etc.
- **Non-sensitive Value:** Attributes that can be directly exposed.

When publishing a dataset, we should remove or obscure the EIDs, hide or blur the QIDs, delete or mask Sensitive Values. All columns except Non-sensitive Values need to be Anonymized. Columns with consistent QIDs after data anonymization is called an equivalence class. If there are multiple records in an equivalence class, users' privacy cannot be accurately obtained through QIDs.

C. Data Desensitization

Data desensitization is mainly used in the development process of software. The general practice is to generate test data, and to require the test data to be basically consistent with the format and scope of the actual data, or to perform certain desensitization on the private data, to make it not expose the user's privacy attributes. The objects of data desensitization include not only private data, but also other sensitive information such as commercial data. It is widely used in the commercial field. Data desensitization mainly achieves desensitization of sensitive data by replacing or deforming sensitive information in the data. Commonly used desensitization technologies mainly include [17]:

- **Substitution:** Replace sensitive data with dummy data, which is usually irreversible.
- **Shuffle:** Change the correspondence between sensitive attribute values and other attributes, so that the privacy attributes cannot accurately correspond to the individual but does not change the statistical characteristics of the privacy attribute.
- **Numerical transformation:** Add random noises within a certain threshold for data of a numerical type.
- **Encryption:** Encrypt sensitive data.
- **Occlusion:** Use '*' to replace part of the sensitive data.
- **Null value insertion or deletion:** Empty sensitive data.

Since the storage of sensitive information in the data table is complicated and multi-correlated in many cases, and the data should be desensitized, the same sensitive information should be kept consistent. Therefore, the data desensitization should also choose different synchronization methods according to the association rules in the table. Traditional static data desensitization does not directly change the data in the database but maintains a mirror library for external access.

D. Application of data protection technology

The commonly used methods are k-anonymity, differential privacy, encryption etc.. Statistical methods such as k-anonymity and differential privacy are to preserve the characteristics of the original data and avoid the association inference, both to ensure data availability and to protect

privacy. The cryptography method mainly focuses on the fact that data is not illegally obtained, but the amount of calculation is generally complicated and is not conducive to further analysis of the data. Therefore, the statistical method is still the mainstream method of privacy protection at present.

Privacy preservation technology is currently receiving more and more attention in the field of genetic data sharing [2], location-based services, Internet of Things, cloud computing, etc. In recent years, the combination of privacy preservation and blockchain [3], the application of differential privacy in deep learning [4] has gradually emerged.

Due to the commercial particularity and sensitivity of data desensitization, many products and patents have been put into commercial use. Alibaba Cloud's data protection umbrella is a product based on the "MaxCompute" platform to protect enterprise data security. Its main functions include sensitive data discovery, sensitive data classification, flexible data desensitization, and abnormal operation monitoring. Its desensitization rules mainly include pseudonyms, hashes, and cover-ups. There are more and more scenarios for data desensitization, and the demand is more and more diversified. How to balance the protection and use of personal information is urgent problems to be solved.

III. PRIVACY PROTECTION DURING THE DATA LIFE CYCLE

Data life cycle includes four stages: data publishing, data storage, data mining and data using [16]. Each stage has a different level of risk on data leakage, so different methods are needed to protect the data during the whole life cycle.

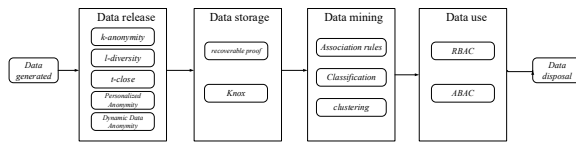


Fig. 1. Privacy Protection During the Data Life Cycle

A. Data release

At the time of data release, the producers or collectors need to ensure both the utility of data and the transparency (or partial transparency) of sensitive data. If only the explicit identifier is removed, it is easy for the adversary to maliciously use the correlation attack and the background knowledge attack.

1) *k-anonymity*

k-anonymity requires that the number of records of any equivalence class in a data table is more than *k*. The degree of privacy protection increases as *k* becomes bigger. It can cover the relationship between the identifier and the sensitive information, leading to a better performance of mitigating the threat caused by correlation attack.

2) *l-diversity*

k-anonymity model can prevent correlation attacks to some extent, but it is useless for the homogeneity attack and background knowledge attack. *l-diversity* model [5] was developed from the *k-anonymity* model. It requires not only there are at least *k* records in an equivalence class but also the value of the sensitive attributes in each equivalence class is more than one. This can effectively prevent background knowledge attack and homogeneity attack. But in some cases, such as the report of disease, there are only two kinds of

indicators—negative and positive, the proportion of the positive is low, but it is very sensitive. Considering this case, the application of 2-diversity may lead to skewed attack and similarity attack.

3) *t-close*

t-close model [6] based on the *l-diversity* takes the distribution of sensitive attribute values of equivalence class into account. This model requires the distribution of sensitive information of equivalence class should be substantially consistent with the overall distribution of sensitive attributes. The distance can't exceed *t*. The EMD (earth mover's distance) is used to measure the distance of the two distribution. *t-close* can effectively prevent skewed type attack and similarity attack but because of the change of the distribution of sensitive attributes, the utility of data is not satisfying. The need to calculate the overall distribution leads to high time complexity.

4) *Personalized Anonymity*

Several methods described above are to change the released table data, achieving the protection of the granularity of the table. If there are multiple sensitive attributes requiring different levels of privacy protection, such as phone numbers and wages, the above method applying the same granularity protection for all sensitive attributes will be useless. Personalized anonymity [7] requires different granularity of privacy protection for different sensitive attributes so that the amount of calculation can be reduced, and the utility of information can be increased.

5) *Dynamic Data Anonymity*

All data anonymization methods above are directed to carry out static data privacy protection. In fact, many of data are dynamically generated. Because the static methods above must know the global data, it is not effective for dynamic data protection. In addition, those methods are easy to be reasoned by adversaries, resulting in privacy leaking. Existing dynamic data privacy protection methods are to protect data by adding, deleting, changing and searching operations. *M-invariance* [8] provides a new way to support dynamic data. It ensures that there are always *m* different values during the whole data releasing process, and the same record has the same value at different times, so the adversary could not apply reasoning attack.

B. Data Storage

Nowadays the data of general enterprises are stored on the cloud server. There are multiple data stored on the third-party data hosting platform. The owner of big data is different from the host of big data, leading to the problem of an untrustworthy third party. The general work combines encrypted storage and third-party audits. The homomorphic encryption algorithm can execute encryption and decryption efficiently in a distributed system and it can also be combined with MapReduce to greatly improve the operation efficiency. Cloud storage audit can ensure that private data will not be changed by the cloud service provider. *POR* [10] (recoverable proof) using an error correction code and the message authentication mechanisms to protect the integrity and recoverability of the data. *Knox* [9] uses group signatures to perform homomorphic authentication, which allows data auditing to be completed without requiring all data.

C. Data mining

Data mining finds the value of the data, but during the data mining process, sensitive properties even anonymized can be

obtained by reasoning attack. Transforming or hiding [11] can achieve privacy protection based on association rules. Random disturbance [12] is used to achieve privacy protection based on the classification result. Translation and flipping [13] and other geometric variations are used to achieve privacy protection based on the results of clustering.

The adversary can often infer the user's private data through some public data of machine learning classifier. Jia [14] proposed a way to add random noise vectors to the public data to reduce the adversary's prediction accuracy. It is mainly based on game theory and heuristic methods to add noise. However, this method is based on the way of predicting the distribution of the adversary. The efficiency and the performance of those methods need to be improved.

D. Data use

Data users may not have the qualifications to obtain the permissions to view or modify data, so in the course of the data using privacy protection also needs to be carried out. For different users, because of their different identities, the availability of data varies, so it is necessary to control the access of users.

Role-based access control (RBAC) [15] is proposed to assign different roles to users, and each role is given different authority to implement resource access control. RBAC emphasizes the strong correlation between permissions and roles. User as a separate entity may have multi-role. How to define the role and authority of one user efficiently based on application scenarios is an urgent problem to be solved.

Attribute-based access control (ABAC) will combine all types of property for user access control, considering a full range of attributes, it can acquire more fine-grained control of access permissions, enabling privacy protection in the process of data access.

IV. DESIGN AND IMPLEMENTATION

This section introduces a way based on the technologies of data anonymization and desensitization for data mining and data privacy threats faced in the course of data using.

A. Experimental Background

In the actual product development, desensitization is also an indispensable part of privacy protection. Data desensitization mainly achieves desensitization of sensitive data by replacing or deforming sensitive information in the data. Current desensitization techniques are primarily for static structured data. For non-structured data, the efficiency of identifying and desensitization is low. And different sensitivity properties should take a different strategy of desensitization.

This experiment wants to deal with structured and unstructured data, applying different and customized strategies to different kinds of sensitive data to protect privacy effectively.

B. Model

The prototype of the system for this experiment is shown in the figure. The system can deal with structured data and unstructured data differently.

For structured data, firstly, which attribute should be privately protected and which level of security is required should be chosen manually. Generally, the secret level can be classified as top secret, confidential, and public. For different

security classification, we may adopt different privacy policies, then anonymize the data. During the test, anonymous data is desensitized.

For unstructured text, sensitive properties need to be pointed out before desensitization in text data, using regular expression and named entity recognition. Then, according to the identified information type and sensitivity level, a differentiated desensitization algorithm is adopted for processing.

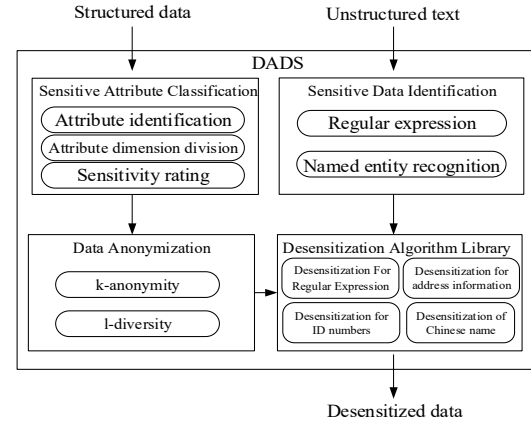


Fig. 2. DADS model based on privacy protection technology

C. Module Design and Implementation

1) Sensitive Attribute Classification

For structured data, firstly, we should determine which columns are EIDs, and which columns are QIDs, then removing the EIDs (data desensitization testing does not need to remove the property).

In different scenarios, sensitive data may have different dimensions. Each dimension corresponds to a different sensitive property. In our system, the security level of an attribute is defined as low, medium, and high.

Different strategies are adopted for different dimensions and levels of data. For high sensitive properties, the consistency of different tables and different rows should be kept after desensitization, and its relevance for other fields. For example, the first 6 numbers of ID number are corresponding with address, the consistency between them should be kept. For medium sensitive attributes, desensitizing can be directly performed. The association and synchronization between the tables could be ignored. For low sensitive properties that are publicly available, there is no need to process.

2) Data Anonymization

Due to the limited computing resources and data, we use k-anonymity to anonymize data in our system. k-anonymity generalizes or suppresses QIDs to achieve that there are at least k records in an equivalence class while ensuring the authenticity and consistency of data before and after the data releasing. The generalization method generally replaces the values in the original data table with more general values, which can include domain generalization and value generalization. Suppression reduces the generalized dimensions by replacing the original data with the general data, making the data more accurate.

3) Sensitive Data Identification

For unstructured text, we first need to identify which sensitive data we need to protect, such as name, address, company, ID number, phone number etc.. After declaring the properties that need to be desensitized, unstructured text is processed to obtain sensitive data. For sensitive data of different formats, the following methods are often used for identification:

a) *Regular expression*: For the information with normative format and scope like ID number, card number, postal code, telephone number and date of birth can use suitable regular expression library to identify. For formatting information such as geographic location may also use an appropriate regular expression to match up.

b) *Named entity recognition*: If there is no formatted composition for information such as person name, organization name and place name, etc., it cannot be identified by matching regular expressions. Natural language processing may be used to recognize named entity. Bi-LSTM + CRF model is used to Chinese named entity recognition in our system, shown in Figure 3. The system first uses one-hot to represent a text sentence as a dense low-dimensional vector, then uses bilayer LSTM to extract text features. Bidirectional LSTM could ensure efficient acquiring of bidirectional characteristics of sentences. Finally, the CRF layer makes entity recognition compliant with common specifications. This method has made the optimal results in named entity recognition areas currently. After identifying the person name, place name and organization name that need to desensitize, the appropriate desensitization rules such as replacement and generalization, etc. could be used for data desensitization.

4) Desensitization Algorithm Library

After confirming which sensitive data or fields requires desensitization, different suitable desensitization methods would be applied to different data fields. This article will use the following data desensitization methods to deal with different categories of information requiring desensitization.

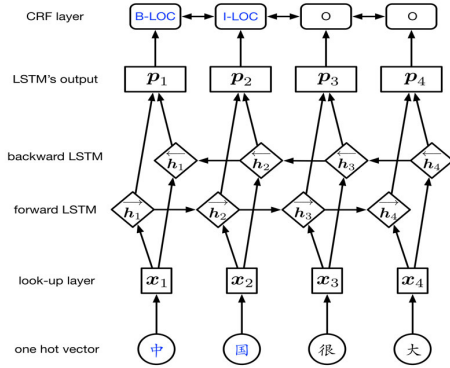


Fig. 3. BI-LSTM + CRF model

a) Desensitization based on the regular expression

Data like phone number, date of birth, mail, can be desensitized by the shutter, a hash function or pseudonym. The default way is to block an intermediate portion of the data field to apply desensitization, e.g., the mobile phone number 13100001234 will be desensitized to 130****1234. In the process of null value insertion, the dummy data is also generated according to the regular expression.

b) Desensitization for ID numbers

Data like ID number and bank card number with strict composition methods can be desensitized by the shutter, pseudonym and hash function. However, when inserting nulls or alternatively changing the real data, the dummy data must be generated in accordance with the norm of data itself, not the regular expression. According to the coding information of ID number, combining public information, we could know: area code has a coding table; date of birth can be generated by regular expressions; the sequence code could be randomly generated and the check code is calculated by weighting through the first seventeen bits of ID number. The generating principle of the bank card number is similar. If the secret level is top secret, it is necessary to ensure the consistency of the dummy data replaced by the same ID number. For this, a unified mapping table can be used for reference.

The dummy data could be an alternative value of the true data and the dummy data need to be blocked under normal circumstances. The desensitization result of the general ID number is shown in Fig.4. As the figure tells us, usable dummy data can be used to replace the true one. And part of the information can be blocked for multi-layer desensitization.

Original ID	Desensitized ID	Masked ID
532525201905164000	44051220181219760X	440512*****760X
130181200502242640	360702200502081525	360702*****1525
610126194301017115	360735194202116000	360735*****6473

Fig. 4. The result of ID number desensitization

c) Desensitization for address information

For location-sensitive information, a desensitization method based on geographic location jitter can be used. Map the actual location information to latitude and longitude space, and disturb the longitude and latitude information, and then analyze the disturbed data to obtain the new and desensitized specific location data. In this process, the degree of desensitization depends on the different security level of address information.

For location-based information, the generalization could be useful for desensitization, using regular expressions to extract the city area name to replace the real data. However, this method has high requirements for the formatting of the geographical locations. If the information is too colloquial or irregular, the effect of desensitizing may be not ideal.

As shown in Fig. 5, the left side is the original data, the middle is the desensitization data after using the latitude and longitude jitter, and the far right is the generalized desensitized address. It can be found that for addresses with clear and unambiguous targets, desensitization with latitude and longitude jitter can be maximally like the true data. The range of jitter could be controlled. In our experiment, it is adjusted to a smaller value to ensure the results in the municipal administrative area. However, for example, the unformatted address like Oriental Pearl, the mapping of latitude and longitude may occur errors and the regular expression is also unsuccessful. We can directly output this value without desensitization.

Original Address	After latitude and longitude jitter	Generalized desensitization
北京市海淀区闵庄路甲89号	北京市房山区佛子庄乡	北京市海淀区
北京市东城区东华门街道天安门	北京市通州区漷县镇草北路	北京市东城区
北京市朝阳区鸟巢	北京市顺义区南彩镇白马路辅路	北京市朝阳区

Fig. 5. Desensitization based on the jitter of geographic location

d) Desensitization of Chinese name

Chinese name desensitization often uses a pseudonym, in the form of popularization of "Bob" or "Lee" instead of the actual name "John Doe." In this experiment, we implemented two methods, one is the "surname + name" replaced by "a name + X", such as "Li Tian" converted to "Li X". The second method is to generate a dummy name and the name should be as consistent as possible with the real name. Thus, using a name library, a statistics corpus containing 120 million common names, set the first and last name weights respectively. The information is stored by the B+ tree. And generates a float point values between 0 and 1, then according to the random number obtained inquire the B+ tree of the first and last name respectively. Finally, we get a full name. This method is of a certain authenticity and independence.

As shown in Figure 6, it produced 15 dummy names, and these names are almost the same as the names we see every day. This method can make it difficult for an adversary to distinguish the authenticity of the information.

Dummy names				
余 兰	闫美花	张雪蕾	时 宾	邓维斗
张 骊	胡华全	张迎春	吴柳钰	谭 子
毛耀锋	郝 帅	刘 薇	林振鹏	李英麟

Fig. 6. Desensitization of Chinese name based on the name corpus

D. The Summary of Experiment

The system not only considers the anonymization of the traditional data table but also the desensitization for unstructured text data. It could protect sensitive information more efficiently. The system is mainly for the stage of data mining and data use, using the data anonymization to defend correlation attacks. And the data desensitization ensures privacy during the process of data using. The system could manage diversity kinds of data and different levels of data with a lot of desensitization methods, having some applicability.

V. CONCLUSION

This paper introduces the basic concept and common methods of privacy protection. DADS was designed and implemented based on those methods. The system combines data anonymization and data desensitization, and it can implement differential desensitization protection strategies for data with various types. This paper mainly studies and realizes the protection technology of static data, and the protection of

sensitive information of dynamic data is the next issue we will discuss.

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